Seyed Alireza Vaezi, PhD

Computer Vision & Machine Learning Scientist

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SUMMARY

Computer Vision & Machine Learning Scientist with a Ph.D. in Computer Science (University of Georgia) who builds reliable segmentation and detection systems where annotated data is scarce. Ph.D. research generated training pseudo-labels with self-supervised and unsupervised methods and adapted foundation vision-language models for biomedical image segmentation. As a postdoc on a USDA-funded pecan processing project, applies computer vision to industrial process optimization: automated video acquisition curates datasets, unsupervised and classical CV methods turn them into pseudo-labels (cutting annotation from thousands of hours to minutes), and the resulting models run on the same line to monitor output quality, laying the groundwork for closed-loop automated process control.

SKILLS

Computer Vision & ML: Python, PyTorch (Lightning), GPU training & inference (PyTorch), Ultralytics YOLO, U-Net, SegFormer, SAM / SAM 2, OpenCV, scikit-image, image segmentation, object detection, tracking, optical flow, self-/unsupervised learning, foundation vision-language models, TensorFlow/Keras

Data & Annotation: NumPy, Pandas, SciPy, Matplotlib, Napari (plugin development), PyQt, dataset curation and augmentation

Deployment & Systems: REST APIs (FastAPI), MQTT, WebSockets, MediaMTX, Supabase, Home Assistant, Docker, Kubernetes, Linux, Git, CI/CD

Vision Hardware & Edge: ESP32, Arduino, ROS, DimOS

Also: C/C++, MATLAB, JavaScript/TypeScript, React

EXPERIENCE

Postdoctoral Research Associate | College of Engineering, University of Georgia **Jun 2025 – Present**

USDA Pecan Post-Harvest Control and Optimization (funded by the U.S. Department of Agriculture)

- **Automated Data Acquisition:** Integrated event-triggered video acquisition into an existing IIoT-instrumented processing line, building a FastAPI service that subscribes to MQTT machine-state events and records RTSP camera streams (via MediaMTX) to capture and log each cracking and shelling run.
- **Computer Vision Integration:** Integrated the cameras with high-intensity lighting for in-line inspection of pecan cracking and shelling, supplying the data pipeline for the team's YOLO model that identifies intact halves, kernel pieces, shell pieces, and occluded views and reports the half-to-full yield ratio as a live process metric.
- **Model Performance & Deployment:** Deployed live YOLO and cascaded U-Net segmentation overlays on production camera streams, reaching over 90% accuracy for each class (shell, kernel, crack region) at 8–12 FPS. The in-line vision integration, live yield estimation, and crack segmentation/classification methods are the subject of an invention disclosure submitted to UGA.
- **Data & Annotation Tooling:** Built a dataset of ~200 processing-run videos (~7 minutes each) and an in-house Napari-based annotation and pseudo-labeling tool with in-viewer training and inference for YOLO, U-Net, cascaded U-Net, SegFormer, and SAM 2, contrastive patch embeddings, instance tracking, and recordable batch pipelines usable without ML expertise.
- **Pseudo-Labeling Pipelines:** Designed the tool's pseudo-labeling workflow, which chains classical image processing (HSV color thresholding, edge-preserving blur, histogram/levels adjustments, edge detection, optical flow/motion analysis, mask editing) into pipelines that transfer to visually similar videos; for a ~1,500-video dataset (~300 frames each), cut estimated manual annotation from ~15,000 hours to ~30 minutes of curating a pipeline on one video and applying it to the rest.
- **Project Leadership:** Coordinate the interdisciplinary USDA pecan project team of 10 graduate students, leading weekly meetings and tracking technical milestones; mentor 5 Ph.D. students and 2 master's students across robotic navigation, generative CAD, 3D vision, and engineering experiment design; lead drafting of federal and industry-partner grant proposals.
- **Industry Engagement:** Conducted site visits with industry partners Southwire, Boehringer Ingelheim, and Label Source to align research with real-world deployment.

Industry Training & Knowledge Transfer

Lead Instructor — "Computer Vision: From Pixels to Practical Applications"
UGA Manufacturing Living Lab Pop-Up Course (Price Industries Innovation Factory — Nov 18, 2025)
Delivered a 2.5-hour hands-on workshop for 21 participants from Price Industries, Boehringer Ingelheim, King's Hawaiian, Kubota, Georgia Power, Southwire, MP Equipment, and K–12 education, providing each participant with an embedded vision kit.

Presenter — Pecan Post-Harvest Research Workshop
UGA College of Engineering (Athens, GA — Aug 19–20, 2026)
Presented computer vision solutions for in-line inspection of pecan cracking and shelling and led an interactive lab tour for nearly 120 attendees. Featured in Pecan South Magazine (Oct 2026) and presented at the Southeastern Pecan Growers Association (SEPGA) Convention.

Graduate Research & Teaching Assistant | University of Georgia **2019 – 2025**

- Served as Graduate Teaching Assistant (2019–2021) supporting computer science courses before transitioning to full-time Graduate Research Assistant (2021–2025) in the Quinn Research Group.
- **Label-Efficient Segmentation:** Trained a supervised cilia segmentation model on self-supervised pseudo-labels instead of manual annotation (SciPy 2024).
- **Foundation Models:** Adapted foundation vision-language models toward a single universal biomedical image segmentation model (Ph.D. dissertation).
- **Cell Tracking:** Built a pipeline for 3D cell instance segmentation, tracking, and motility classification of Toxoplasma gondii (SciPy 2022).
- **Microscopy Analysis:** Developed a computational model of mitochondrial morphology changes caused by Mycobacterium marinum infection and deep-learning-assisted detection of mitochondrial fusion and fission in microscopy videos.
- **Human-Centered Industrial CV:** Developed a non-intrusive emotion recognition framework using facial expression and body posture to assess worker stress, fatigue, and cognitive load in IIoT-enabled manufacturing, in collaboration with Dr. Jaime Camelio's lab (NAMRC 52).
- **Open-Source Tooling:** Developed TSeg, an open-source Napari plugin that packages off-the-shelf segmentation algorithms for microscopy cell image analysis by non-experts.

SELECTED PUBLICATIONS

- **Vaezi, S. A.**, et al. "Training a Supervised Cilia Segmentation Model from Self-Supervision." Proc. 23rd Python in Science Conference (SciPy 2024).
- **Vaezi, S. A.**, et al. "A Novel Pipeline for Cell Instance Segmentation, Tracking and Motility Classification of Toxoplasma gondii in 3D Space." Proc. 21st Python in Science Conference (SciPy 2022).
- Rezaei, N., **Vaezi, S. A.**, Camelio, J. "Decoding the Silent Language: A Framework for Advanced Visual Emotion Recognition to Enhance Worker Well-Being on IIoT-Enabled Manufacturing." NAMRC 52 (2024).
- Rezaei, N., **Vaezi, S. A.**, et al. "Smart Sensors, Smarter Workplaces: Multi-Modal Cognitive Load Assessment in Industry 4.0." IISE Annual Conference (2025).
- Quinn, S., Abbadi, A., **Vaezi, S. A.** "Identifying Virulence Determinants in Pathogenic Mycobacteria via Changes in Host Cell Mitochondrial Morphology." arXiv.

EDUCATION

Ph.D., Computer Science | University of Georgia, Athens, GA | Jan 2019 – May 2025 | GPA: 3.74
Dissertation: "From Specific to Universal: One Biomedical Image Segmentation Model to Rule Them All"

M.S., Information Technology Engineering (Computer Networks) | Iran University of Science and Technology, Tehran, Iran | 2016

B.S., Information Technology Engineering | Azad University of Parand, Tehran, Iran | 2012